AF5638 Final Practice Questions

(Dynamically Updated)

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1. Key Concepts and knowledge points

Topic 6 – Fiscal policy: Ch. 13

- Government intertemporal budge constraint
- Ricardian equivalence
- Government spending multiplier
- Crowding out

Topic 7 – Equilibrium in Production Economy: Ch. 12 (excluding 12.1.1)

- Conceptually understand the problems faced by households and firms: their objectives, the choices they make, and the variables they take as given
- Be able to describe the economic intuition behind each first-order condition that governs the optimal choices of economic agents (Note: the first-order conditions will be provided on the exam; you are not required to derive them)
- Be able to determine the sign of the partial derivatives for the consumption function, labor supply function, investment demand function, and labor demand function, and explain the economic intuition behind each
- Understand the labor market and goods market clearing conditions, which ensures that the funds market also clears
- Be able to identify which variables are endogenous and which are exogenous in this environment

Topic 8 – Neoclassical Model (Real Block): Ch. 18, 19

- Ch. 18 covers the graphical representation of the equilibrium from Topic 7
- Ch. 19 uses graphs to analyze how the neoclassical economy responds to different shocks (ignoring the nominal block for now)
- Understand how the IS and Y^s curves are derived and how equilibrium is established
- Be able to explain how various shocks shift either the IS or Y^s curves and the resulting general equilibrium effects on endogenous variables

2. [Fiscal Shocks, 2025 Final] In the Neoclassical model, aggregate desired expenditure is given by:

$$Y_t^d = C^d(Y_t - G_t, Y_{t+1} - G_{t+1}, r_t) + I^d(r_t, A_{t+1}, K_t) + G_t.$$

Market clearing requires that $Y_t = Y_t^d$.

a) Briefly explain what Ricardian equivalence entails.

b) In the version of the model presented above, is Ricardian equivalence assumed to hold? Briefly explain.

c) What is the sign of the partial derivative of the consumption function $C^d(\cdot)$ with respect to the first two arguments (i.e., $Y_t - G_t$ and $Y_{t+1} - G_{t+1}$)? Use a hypothesis we learned in this class to provide a direct answer.

d) The investment demand function $I^d(\cdot)$ is derived from the optimality condition governing the firm's optimal choice of K_{t+1}, which is:

$$1 + r_t = A_{t+1} \frac{\partial F(K_{t+1}, N_{t+1})}{\partial K_{t+1}} + (1 - \delta).$$

Intuitively describe what each side the optimality condition captures and why they are equalized at optimum.

[Recall in the model we have assumed that the representative firm owns the capital stock in the economy, which evolves according to $K_{t+1} = I_t + (1 - \delta)K_t$, and the firm needs to borrow from the bank to finance I_t at a real interest rate r_t, paid back to the bank in period $t + 1$.]

e) Suppose output is already determined by the supply side at a level of $\overline{Y_t}$. On a graph with r_t on the vertical axis and Y_t on the horizontal axis. Draw the IS curve and the Y^s curve that determine the equilibrium real interest rate.

f) On another graph with Y_t^d on the vertical axis and Y_t on the horizontal axis, draw the aggregate desired expenditure line and the 45-degree line that is consistent with the equilibrium.

g) Show what happens to the graph in part (e) and the graph in part (f) following an exogenous increase in G_t. Summarize the effect on equilibrium real interest rate and equilibrium output.

h) Relate your answers to part (g) to a key implication of the Neoclassical model.

3. [Neoclassical Model Real Block] Consider a two-period production economy. There is no government spending. The household has optimal decision rules:

$$C_t = C^d(Y_t, Y_{t+1}, r_t)$$
$$N_t = N^s(w_t, \theta_t)$$

A representative firm produces output according to the production function $Y_t = A_t F(K_t, N_t)$. The production function has the standard properties. The optimal decision rules for the firm are:

$$N_t = N^d(w_t, A_t, K_t)$$
$$I_t = I^d(r_t, A_{t+1}, K_t)$$

Market clearing requires that $Y_t = C_t + I_t$.

a) Specify the endogenous variables and exogenous variables in this model.

b) List the set of equations characterizing the equilibrium.

c) Sign the partial derivative of the household's labor supply function with respect to each argument, and provide an intuitive explanation (θ_t is the preference parameter for leisure as defined in class).

d) The labor supply function is derived from the following first-order condition:

$$\frac{\partial u(C_t, L_t)}{\partial L_t} = w_t \frac{\partial u(C_t, L_t)}{\partial C_t},$$

where $L_t = 1 - N_t$ denotes leisure. Explain what each side of the equation captures and why they should be equal to each other at the optimum.

e) Draw in the main graphical diagram for this model to show the initial equilibrium.

f) Suppose that A_t increases (A_{t+1} remains unchanged). In your graphical diagram above, show how this shock impacts all relevant curves and carefully label and new equilibrium. Indicate whether each endogenous variable goes up, down, is unaffected, or if the effect is ambiguous.

g) Relative to an endowment economy where $Y_t = A_t$, does output Y_t increase by more or less in the production economy when A_t increases? Briefly explain your answer.

h) How does consumption adjust to higher A_t relative to Y_t? How does your answer here differ (or not) from an endowment economy setup? Explain briefly.